

# The biological role of sensory responses variability

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Neuronal responses to sensory stimulation, in primary sensory brain regions, can be extremely variable even when the presented stimulus keeps the same physical properties. This variance can be represented in the level of the response of single neurons but also at the number of neurons involved in the response. This variance in the sensory response could represent noise of the circuitry, or it could carry important information such as the internal state of the circuit or animal -attentional processes, level of hunger, etc.

In the zebrafish larval model, two-photon calcium imaging experiments have shown that when stimulus consist in moving bars that slice space in a given direction at a given speed, and no more fixed dots that would allow to draw receptive fields, neuron responses are highly variable. This is a pretty interesting observation to make, since noise correlation (trial to trial variability for a fixed stimulus) and signal correlation (correlations in the average response) do not systematically enhance information carrying in a network: so maybe this variability could participate to an optimal encoding of the stimulus. Such an assumption, crossing network dynamics study and information theory, has been proposed and validated in A.Pouget et al's work in mouse visual cortex (information limiting correlations).

Zebrafish is a good vertebrate to study such problematic since it is a small organism whose brain activity can easily be recorded thanks to optogenetics: at larva state, it is transparent, Gcamp lines are available for optogenetics so brain recording can directly be implemented. What's more retinotopically-organized visual information processing in retinal and post retinal neurons has widely been studied, so such a work could bene t from well established basic knowledge. What's more, the relatively small number of neurons describing the system make a network model validation easier in terms of identi cation arti cial neuron/real neuron for example (compared to human brain for example). A network model reproducing observed tuning curves and correlation patterns could even be proposed of a well triggered zone.

To study this important question of neural activity variability, fundamental in neuroscience, we will use a multidisciplinary approach combining calcium imaging, optogenetics and a proposition of mathematical modeling for optic tectum visual information processing.

## 1. Biological materials

The proposed experimental framework is to study zebrafish optic tectum processing of visual stimuli, to understand how the variability previously observed in evoked responses could take place in a global encoding scheme. Stimuli that could be presented can be split into two categories:

- fix dots stimulus, that allows to draw receptive fields structure,
- moving bar stimulus, that could permit to understand how features like direction or velocity could be encoded.

First preliminary decoding approaches and variance studies have been implemented during the two past months, that underly the need for a larger number of different stimulus (the experiment used for the preliminary analysis present only four directions) and the need for an increased temporal resolution. Indeed, tuning an encoding model with observed neural activity that would shape its variability necessitates to have a spike-level time precision for the recorded data, so that reliable tuning curves or statistics on spike trains can be computed without the well-known problems that canonical deconvolution tools can present with  $ca2+$  imaging data.

So, technically speaking, the key exigence would be to be able to get a temporal resolution high enough to trigger spikes or ring rates. Two solutions exist to achieve such a requirement: either using gcamp fishes and spim imaging at 100z and implementing a deconvolution (thanks to existing SPIM set-up), either getting directly the membrane potential thanks to newly developed genetically encoded voltage indicators (Ace1Q-mNeon [5],[6]).

## 2. Response variability analysis

I started a data-analysis work on neural activity in response to basic visual stimuli, thanks to an experimental protocol implemented by Jonathan Boulanger-Weill (IBENS, post-doctorant) [9].

Two sets of stimuli were presented: - Bars in each 4 cardinal directions, - Squares of  $5^\circ$  length. Larva's brain activity was recorded thanks to two-photon microscopy.

Optic terminals are distributed in an orderly fashion over tectum so as to make a map of the retina. The retinotectal map is organized so that the front, back, top and bottom of the visual field seen by one eye is represented at the front, back, top and bottom of the opposite tectal lobe. Receptive fields sizes are about 25-39 degrees. Such a retinotopical map induces a strong patterns structure in correlations.

The first step is to explore how signal correlations and noise correlations are impacted by this nervous anatomy.

What's more, given that I tried several decoding approaches to see how the stimulus could be deduced from neural activity using methods described in [4], the question that remains to be explored is to see if there is a link between neuronal response's variability and decoding performance, to have an empirical intuition of how variability and information can be linked.

### 3. Modeling

Boerling, Machens, Deneve et al [1] have recently proposed a spiking network model able to track or integrate a stimulus, where optimal efficiency is reached with balance between inhibitory and excitatory input connections for a given neuron. The input of the network is a describing variable that has its proper dynamics written as a function of the stimulus. This network is composed of leaky integrate and re neurons (LIF neurons), that all together are able to implement a linear system minimizing a loss function. The loss function consist in the difference between the describing variable and its approximation by the network, plus some constraint that enhance the network to spike sparsely with a repartition of the metabolic cost all over the population. The decoded variable -the output- is a direct linear combination of the spike trains; more precisely, the idea is that the relevant information can be extracted by postsynaptic integration.

Such a model predicts a detailed balance between excitation and inhibition, without imposing it upfront. This balance follows from the attempt of the network to minimize the loss function, which in turn implies that the membrane potential of neurons represents a prediction error and that neurons spike only when this prediction error exceeds a certain value. The network connectivity and spike generation are entirely derived from a single loss function. This approach implies in particular that neurons share information in a smart way, yet re almost randomly at the level of the single cell.

This model could be applied to the zebrafish visual system, even if the balance evoked here remains to confirm more precisely biologically. Indeed, the variability of the sensory response in one hand, and this balance on the other hand, are direct consequences of such a spiking network model when optimized to efficiently represent information. The network is described by 3 parameters: the number of neurons, the kernels that delimit subspaces each neuron is going to act on more specifically, and network dynamics that describes ring rate profile's update given a new spike.

Even if this network could be used as a single machine learning tool with interesting correspondences or analogies with biological functionalities, one should not forget that most of the work describes in the paper consists in showing how close to membrane potential equations or effective connectivities such a network can be, with well chosen variables. Indeed, such parameters allow us to decide if the network tracks or integrates sensory information, fix the intensity of slow connexions and fast connexions (this two ways decomposition of correlation can lead to an interesting biological exploration of levels of dynamics in tectum neural circuits), and allow the network to reduce encoding error by producing one or several spikes at each time.

Such a greedy correction is interesting since it allows a continuous adaptation to stimulus, and seems well fitted to described adaptation latency one can observe of 'habituation' phenomenon. What's more, having as an entry not only a stimulus but also its neural representation, such a modelization allows for the exploration of the notion of short term memory or work memory one could want to understand in neurobiological systems.

The tuning and the validation of the model is not trivial: the simple use of the word 'validate' imply an epistemological view of what can decide or not of the relevance of a model. The fact that it could t the dynamics experimentally observed is not sufficient or convincing enough. Plausibility must be worked at a theoretical level working on the representation of the neurons and the connections we chose to write. That's why modelization work and data analysis can't be decorrelated in this work. Therefore, there will be two main axes with goings and comings from one to the other:

A – We will first ask if the network model such as written today can be tested as plausible enough.

As this specific model has been proposed for the cortex without taking into account the possibility of schemes of co-activation assemblies, and as co-activation assemblies has been observed in optic tectum (Romano et al, 2015 [8]). one idea could be to replace the sparsity constraint by a group-sparsity constraint. Such a work consists in rewriting the cost function of the model.

Another modification could be to implement a more than one layer dynamical tracking, since the model is composed by a single layer as it has been presented and implemented up to now. But such a modification is not trivial, and should be drawn carefully. It would represent a great amount of work and should not be understood as a final aim since the relevance of this idea still remains to be validated.

B – As a validation process, we will specifically ask:

1) Whether there are optimal network's parameters that can be learnt from neural data-to-stimulus decoding, more precisely from correlational structures like time by time correlation matrices. We know from literature and works on mouse visual cortex that information depends on both signal correlation and noise correlation, and that the existence of strong patterns of correlation can limit total information in the network (J.Beck, R.Moreno-Bote, 2014 [2]). Since this observation has been made experimentally, and since the existence of information limiting correlations in case of strong correlations patterns has been demonstrated mathematically, we should wonder what kind of correlations should be used to deduce the kernels of the artificial network proposed as a functional model for zebra sh optic tectum studying.

2) Whether, with such a set of parameters, the artificial network can presents tuning curves and performances scales close to biological ones. Is there an identification LIF-neuron/biological neuron possible? Or should our abstract neural entities represent linear combination of neurons co- varying together for example? This question is central and won't come to our mind at the late end of the work, since it shapes the whole heuristic routine of the problematic we propose here.

## References

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